Heat is not temperature

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish temperature from energy transferred because of a temperature difference.

SUCCESS CRITERIA - I can...

1. Compare temperature with thermal energy.
2. Identify the direction of spontaneous heat transfer.
3. Explain why amount and material matter.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “heat” and “temperature” to explain your answer. Sentence starters are optional.

heat: energy transferred because of a temperature difference

temperature: a measure of thermal state; it is not an energy amount

thermal energy: internal energy associated with microscopic motion/interactions

01 Warm-up: bring back an earlier idea

For fixed GM, launch radius becomes **9** times as large. **Compare ideal escape speeds** and **name the equation**.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $Q = mc\,\Delta T$.
For heating without a phase change: Q is transferred heat, m is mass, c is specific heat. Heat and temperature are different quantities.

Worked reasoning: A cup and a bathtub can both be at **40°C**. Equal temperature does not imply equal internal energy: amount and type of material also matter.

Your turn: 1. **Draw equal temperature indicators but** different amounts of water.
2. **Predict heat-flow direction** if the water samples are brought into thermal contact at the same temperature.

Sentence starter: Both ... . They differ in ... because ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A cup and a bathtub contain the same substance at **40°C**. Are their temperatures equal? Must their total internal energies be equal?

Sentence starter: I notice ... . This supports ... because ... .





Cup: 40°C

Tub: 40°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

Two containers hold the same liquid at the same temperature. Container B has three times as much liquid by mass.
1. **Compare the energy needed** to warm each by the same temperature change.
2. **Label the diagram** and say what you are treating as true.

Sentence starter: Both … . They differ in … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student says “cold flows into a hot drink.” Rewrite the explanation in terms of energy transfer.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A point-mass pendulum drops **0.8 m** vertically from rest. Use g=**10** and neglect drag. **Find bottom speed**. **Show the energy changes** and transfers; **label units**.

Word help: pendulum = an object that swings about a fixed support

Sentence starter: I found … [unit]. This value means … in this situation.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

Can equal-temperature objects still contain different internal energies? **Give two reasons**.

Sentence starter: Both … . They differ in … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two equal masses of water start at **20°C** and **60°C** and mix in isolation without phase change. **Find final temperature**, then **explain why this averaging rule** would fail for unequal masses.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3